

FinRelief AI — Feature Overview

SmartBridge Internship Program

Version 2.0.0

1. Pages

Route	Page	Description
/login	Login	Email/password login, plus a one-click demo account for exploring the app without a real login
/register	Register	New account creation
/dashboard	Dashboard	Aggregate KPIs, debt stress trend charts, and negotiation letter history
/loans	Loans	Full create/read/update/delete management of a user's loans
/settlement	Settlement	Runs a settlement simulation against a selected loan
/letters	Letters	Generates, edits, and downloads OTS negotiation letters as PDFs

2. Functional Features

Authentication

Registration and login with JWT-based sessions. Passwords are bcrypt-hashed. A one-click demo login is available so a new visitor can explore the full app immediately without creating an account or entering real financial information.

Loan Management

Full CRUD on loans — lender name, loan amount, EMI, days overdue, and monthly income. These four fields are the entire input surface the rest of the app's calculations are built from.

Settlement Simulation

Given a loan's stored data, the backend computes a debt-to-income ratio, a debt stress level, and a recommended One Time Settlement percentage. Each run is saved as a snapshot, which is what powers the dashboard's trend charts — a user can see their stress level improve or worsen over time, not just at a single point.

AI Negotiation Letter Generation

Turns a settlement calculation into an actual, professionally worded letter addressed to the specific lender, using Google Gemini 2.0 Flash. If Gemini is unavailable, a static professional template is used instead automatically — the feature never simply fails. Letters can be edited after generation and downloaded as a PDF via ReportLab. A full version history is kept per loan.

Financial Health Dashboard

Aggregates KPIs across all of a user's loans: current stress levels, DTI trends over time, and a history of every generated negotiation letter, so the whole financial picture is visible in one place.

3. Non-Functional Characteristics

Usability — The three core scenarios (get a settlement number, generate a letter, track progress) are each reachable in a small number of clicks from the dashboard, which is treated as the home base of the app.

Security — Bcrypt password hashing, JWT-based stateless sessions, environment-variable-only secrets, and SlowAPI rate limiting on sensitive endpoints.

Reliability — The Gemini fallback pattern is the single most important reliability decision in the app: letter generation cannot be broken by an AI service outage, because there is always a deterministic fallback path.

Performance — Settlement calculations are synchronous, pure functions with no external calls and return effectively immediately. The only externally-dependent, latency-sensitive call in the system is the Gemini request during letter generation.

Scalability — The decoupled frontend/backend architecture, combined with stateless JWT auth, means the backend can be scaled horizontally without any shared session state to coordinate across instances.

4. Development Milestones

1. **Environment Setup & Dependency Configuration** — Python 3.11+, React + Vite, FastAPI, SQLite & SQLAlchemy for local/test use, Google Gemini API integration
2. **AI Integration & Financial Processing** — Debt stress calculation, DTI analysis, settlement percentage prediction, Gemini-driven negotiation letter generation
3. **Core Backend Development** — Authentication, user management, loan management, financial analysis, and settlement prediction as FastAPI services
4. **Database Management & Persistence** — Users, loans, settlement snapshots, and AI-generated letter history, persisted in Neon PostgreSQL
5. **Frontend UI Development** — Login/registration, dashboard, settlement simulator, and letter generator, built as a responsive React SPA
6. **Testing, Optimisation & Deployment** — Automated test coverage, security hardening, and deployment to Render (backend) and Vercel (frontend)

5. Project Metrics

Metric	Value
Epics	10

Metric	Value
Stories	26
Tech stack layers	6
API endpoints	21
Test cases	68

6. Team

Name	Role
Sri Datta Sai Vithal Kaligotla	Team Lead
Bhoganaduni Manogna	Team Member
Jayasree Movva	Team Member
Sri Satya Lahari Pragada	Team Member
Krishna Sanjay Pemmaraju	Team Member